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भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Mathematical Foundations of Data Science (Probability)

Lecture 08

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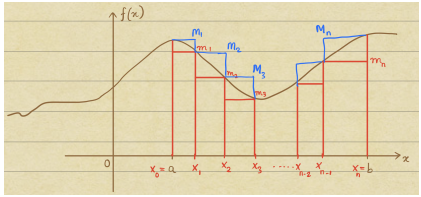
03 October 2026

A Primer on Riemann Integration

Let $f : \mathbb{R} \to \mathbb{R}$ be given.
Let $a, b \in \mathbb{R}$, $a < b$, be given.

What is the mathematical interpretation of $\int_a^b f(t)\, dt$?

A Primer on Riemann Integration



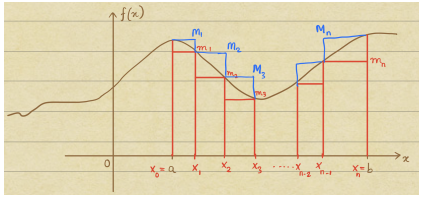
- Create a partition $\mathcal{P}_n = \{x_0, x_1, \ldots, x_n\}$, where $x_0 = a$ and $x_n = b$
- Define the lower Riemann sum as the total area under the red rectangles, i.e.,

$$L(f, \mathcal{P}_n) = \sum_{i=1}^{n} m_i \cdot |x_i - x_{i-1}|.$$

- Define the upper Riemann sum as the total area under the blue rectangles, i.e.,

$$U(f, \mathcal{P}_n) = \sum_{i=1}^{n} M_i \cdot |x_i - x_{i-1}|.$$

A Primer on Riemann Integration



- It is easy to see that for all n,

$$L(f, \mathcal{P}_n) \leq L(f, \mathcal{P}_{n+1}), \qquad U(f, \mathcal{P}_n) \geq U(f, \mathcal{P}_{n+1})$$

- If $\lim_{n\to\infty} L(f, \mathcal{P}_n) = \lim_{n\to\infty} U(f, \mathcal{P}_n)$, then this common limit is denoted $\int_a^b f(t)\, dt$

What if $\lim_{n\to\infty} L(f, \mathcal{P}_n) \neq \lim_{n\to\infty} U(f, \mathcal{P}_n)$?

Consider the function

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

For the above function:

- $L(f, \mathcal{P}_n) = 0$ for all n
- $U(f, \mathcal{P}_n) = 1$ for all n

Remedy:

A more general theory of integration, proposed by Lebesgue!

Lebesgue's Integration Theory

Consider a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let $X : \Omega \to \mathbb{R}$ be a random variable with respect to $\mathscr{F}$.

Expectation

The expectation of a random variable X is defined as

$$\mathbb{E}[X] \;=\; \int_{\Omega} X \, d\mathbb{P}.$$

Furthermore, for any $A \in \mathscr{F}$, we define

$$\mathbb{E}[X\,\mathbf{1}_A] \;=\; \int_{A} X \, d\mathbb{P}.$$

Lebesgue builds the above definition of expectation in 3 stages:

- Definition of the abstract integral for "simple" random variables
- Definition of the abstract integral for non-negative random variables
- Definition of the abstract integral for arbitrary random variables

Properties of Expectations

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

1. If $A \in \mathscr{F}$ such that $\mathbb{P}(A) = 0$, then for any random variable X, we have
$$\mathbb{E}[X \cdot \mathbf{1}_A] \;=\; 0.$$
2. If $A \in \mathscr{F}$ such that $\mathbb{P}(A) = 1$, then for any random variable X, we have
$$\mathbb{E}[X \cdot \mathbf{1}_A] \;=\; \mathbb{E}[X].$$
3. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
4. $\mathbb{E}[a\,X] = a\,\mathbb{E}[X]$ for all $a \in \mathbb{R}$.

Expectations for Discrete and Continuous RVs

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let $X : \Omega \to \mathbb{R}$ be a random variable with respect to $\mathscr{F}$.

- If X is a discrete random variable with PMF p_X, then

$$\mathbb{E}[X] = \sum_x x \cdot p_X(x), \qquad \mathbb{E}[g(X)] = \sum_x g(x) \cdot p_X(x).$$

- If X is a continuous random variable with PDF f_X, then

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x \cdot f_X(x)\, \mathrm{d}x, \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) \cdot f_X(x)\, \mathrm{d}x.$$

Examples

- Let X be a discrete random variable with the PMF

$$p_X(x) = \begin{cases} 0.1, & x = 1, \\ 0.2, & x = -2, \\ 0.2, & x = 3, \\ 0.5, & x = -4, \\ 0, & \text{otherwise.} \end{cases}$$

  Compute $\mathbb{E}[X]$.

Examples

- Let X be a discrete random variable with the PMF

$$p_X(x) = \begin{cases} \dfrac{1}{x(x+1)}, & x \in \mathbb{N}, \\ 0, & \text{otherwise.} \end{cases}$$

Compute $\mathbb{E}[X]$.

Examples

- Let X be a discrete random variable with the PMF

$$p_X(x) = \begin{cases} \frac{3}{\pi^2} \cdot \frac{1}{x^2}, & x \in \mathbb{Z}, \\ 0, & \text{otherwise.} \end{cases}$$

  Compute $\mathbb{E}[X]$.

Examples

- Let X be a continuous random variable with the PDF

$$f_X(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2}, \qquad x \in \mathbb{R}.$$

What is $\mathbb{E}[X]$?

Examples

- Compute the mean of $X \sim \text{Ber}(p)$.
- What is the mean of $X \sim \text{Poisson}(\lambda)$?
- What is the mean of $X \sim \text{Unif}([a, b])$?
- What is the mean of $X \sim \text{Exponential}(\mu)$?
- What is the mean of $X \sim \mathcal{N}(\mu, \sigma^2)$?

On the Value of Expectation

On the Value of Expectation

Expectation of a random variable satisfies one of the following:

- Expectation is a **real number**, i.e., a finite value
- Expectation is **equal to** $+\infty$
- Expectation is **equal to** $-\infty$
- Expectation is **undefined**, i.e., it is of the form $\infty - \infty$

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Variance, Covariance, and Correlation

Variance

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X be random variable with respect to $\mathscr{F}$.
Let $\mathbb{E}[X]$ be well defined (i.e., not of the form $\infty - \infty$).

Definition (Variance)

The variance of X is defined as

$$\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Remarks:

- $\mathrm{Var}(X) \geq 0$.

Variance

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
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Definition (Variance)

The variance of X is defined as

$$\mathrm{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Remarks:

- $\mathrm{Var}(X) \geq 0$.
- The quantity $\sqrt{\mathrm{Var}(X)}$ is called the standard deviation of X.

A Result on Zero Variance

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X be random variable with respect to $\mathscr{F}$.
Let $\mathbb{E}[X]$ be well defined (i.e., not of the form $\infty - \infty$).

Lemma (Zero Variance)

The variance of X is zero if and only

$$\mathbb{P}(\{X = c\}) = 1 \qquad \text{for some constant } c.$$

An Alternative Expression for Variance

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X be random variable with respect to $\mathscr{F}$.

Alternative Expression for Variance

Let $\mathbb{E}[X]$ be well defined (i.e., not of the form $\infty - \infty$).

1. If $\left|\mathbb{E}[X]\right| = +\infty$, then $\mathrm{Var}(X) = +\infty$.
2. If $\left|\mathbb{E}[X]\right| < +\infty$, then

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

Examples

- Compute the variance of $X \sim \text{Ber}(p)$.
- What is the variance of $X \sim \text{Poisson}(\lambda)$?
- What is the variance of $X \sim \text{Unif}([a, b])$?
- What is the variance of $X \sim \text{Exponential}(\mu)$?
- What is the variance of $X \sim \mathcal{N}(\mu, \sigma^2)$?
- Give an example of a random variable X for which $\left|\mathbb{E}[X]\right| < +\infty$, but $\text{Var}(X) = +\infty$.

Covariance

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathscr{F}$.
Let $\mathbb{E}[X], \mathbb{E}[Y]$ be well defined (i.e., not of the form $\infty - \infty$).

Definition (Covariance)

The **covariance** of X and Y is defined as

$$\mathrm{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[\mathbb{Y}])],$$

provided the expectation on the right-hand side is well defined (i.e., not $\infty - \infty$).
Furthermore,

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y],$$

provided the right-hand side is not of the form $\infty - \infty$.

Uncorrelated Random Variables

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathscr{F}$.
Let $\mathbb{E}[X], \mathbb{E}[Y]$ be well defined (i.e., not of the form $\infty - \infty$).

Definition (Uncorrelated Random Variables)

X and Y are said to be uncorrelated if

$$\mathrm{Cov}(X, Y) = 0.$$

Uncorrelatedness and Independence

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathscr{F}$.
Let $\mathbb{E}[X], \mathbb{E}[Y]$ be well defined (i.e., not of the form $\infty - \infty$).

Theorem (Uncorrelatedness and Independence)

If $X \perp\!\!\!\perp Y$, then

$$\mathrm{Cov}(X, Y) = 0.$$

The converse is not true in general. For example, consider

$$X \sim \mathcal{N}(0, 1).$$

Let $Y = X^2$. Then, it is easy to verify that $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$, and $\mathbb{E}[X]\mathbb{E}[Y] = 0$.
Therefore, $\mathrm{Cov}(X, Y) = 0$, but $X \not\perp\!\!\!\perp Y$.

Variance of Sum of Two Random Variables

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathscr{F}$.

Lemma (Variance of Sum of Two Random Variables)

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y).$$

Variance of Sum of Two Random Variables

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathscr{F}$.

Lemma (Variance of Sum of Two Random Variables)

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y).$$

Remark:
If X, Y are uncorrelated, then

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

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Correlation Coefficient and Cauchy–Schwartz Inequality

Correlation Coefficient

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathscr{F}$.

Definition (Correlation Coefficient)

The correlation coefficient of X and Y is defined as

$$\rho_{X,Y} := \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)}}.$$

Correlation Coefficient

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathcal{F}$.

Definition (Correlation Coefficient)

The correlation coefficient of X and Y is defined as

$$\rho_{X,Y} := \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)}}.$$

Remark:
$\rho_{X,Y}$ can be positive, negative, or zero

The Cauchy–Schwartz Inequality

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Theorem (Cauchy–Schwartz Inequality)

For any two random variables X and Y,

$$-1 \leq \rho_{X,Y} \leq 1.$$

Furthermore, the following hold.

1. If $\rho_{X,Y} = 1$, then there exists $a > 0$ such that

$$\mathbb{P}(\{Y - \mathbb{E}[Y] = a\,(X - \mathbb{E}[X])\}) = 1.$$

2. If $\rho_{X,Y} = -1$, then there exists $a < 0$ such that

$$\mathbb{P}(\{Y - \mathbb{E}[Y] = a\,(X - \mathbb{E}[X])\}) = 1.$$

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Conditional Expectations

Conditional Expectation

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X, Y be random variables with respect to $\mathscr{F}$.

Objective

To define the following quantities:

- $\mathbb{E}[X|\{Y = y\}]$, for any $y \in \mathbb{R}$.
- $\mathbb{E}[X|Y]$.

Programme:
We shall define the above quantities by considering X discrete/continuous, and Y discrete/continuous.

Case 1: X, Y Jointly Discrete

Let X, Y have the joint PMF $p_{X,Y}$.

- Step 1: Conditional PMF of X, conditioned on the event $\{Y = y\}$:

$$p_{X|Y=y}(x) = \frac{p_{X,Y}(x, y)}{p_Y(y)}, \qquad x \in \mathbb{R}.$$

- Step 2: The quantity $\mathbb{E}[X|\{Y = y\}]$ is defined as the expectation with respect to the conditional PMF $p_{X|Y=y}$, i.e.,

$$\mathbb{E}[X|\{Y = y\}] := \sum_{x \in \mathbb{R}} x \cdot p_{X|Y=y}(x).$$

Case 1: X, Y Jointly Discrete

Let X, Y have the joint PMF $p_{X,Y}$.

- Step 3: Define the function $\psi_1 : \mathbb{R} \to \mathbb{R}$ as

$$\psi_1(y) := \begin{cases} \mathbb{E}[X|\{Y = y\}], & p_Y(y) > 0, \\ 0, & p_Y(y) = 0. \end{cases}$$

- Step 4: The quantity $\mathbb{E}[X|Y]$ is simply defined as

$$\mathbb{E}[X|Y] = \psi_1(Y).$$

Case 2: X, Y Jointly Continuous

Let X, Y have the joint PDF $f_{X,Y}$.

- Step 1: Conditional PDF of X, conditioned on the event $\{Y = y\}$:

$$f_{X|Y=y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \qquad x \in \mathbb{R}.$$

- Step 2: The quantity $\mathbb{E}[X|\{Y = y\}]$ is defined as the expectation with respect to the conditional PDF $f_{X|Y=y}$, i.e.,

$$\mathbb{E}[X|\{Y = y\}] := \int_{-\infty}^{+\infty} x \cdot f_{X|Y=y}(x).$$

Case 2: X, Y Jointly Continuous

Let X, Y have the joint PDF $f_{X,Y}$.

- Step 3: Define the function $\psi_2 : \mathbb{R} \to \mathbb{R}$ as

$$\psi_2(y) := \begin{cases} \mathbb{E}[X|\{Y = y\}], & f_Y(y) > 0, \\ 0, & f_Y(y) = 0. \end{cases}$$

- Step 4: The quantity $\mathbb{E}[X|Y]$ is simply defined as

$$\mathbb{E}[X|Y] = \psi_2(Y).$$

Case 3: X Continuous, Y Discrete

- Step 1: Conditional CDF of X, conditioned on the event $\{Y = y\}$:

$$F_{X|Y=y}(x) = \mathbb{P}(\{X \leq x\}|\{Y = y\}), \qquad x \in \mathbb{R}.$$

- Step 2: Get the conditional PDF of X, conditioned on the event $\{Y = y\}$:

$$h_y(x) = \frac{d}{dx} F_{X|Y=y}(x).$$

- Step 3: The quantity $\mathbb{E}[X|\{Y = y\}]$ is defined as the expectation with respect to the conditional PDF $h_y(x)$, i.e.,

$$\mathbb{E}[X|\{Y = y\}] := \int_{-\infty}^{+\infty} x \cdot h_y(x)\, dx.$$

Case 3: X Continuous, Y Discrete

- Step 4: Define the function $\psi_3 : \mathbb{R} \to \mathbb{R}$ as

$$\psi_3(y) := \begin{cases} \mathbb{E}[X|\{Y = y\}], & p_Y(y) > 0, \\ 0, & p_Y(y) = 0. \end{cases}$$

- Step 5: The quantity $\mathbb{E}[X|Y]$ is simply defined as

$$\mathbb{E}[X|Y] = \psi_3(Y).$$

Case 4: X Discrete, Y Continuous

- Step 1: The joint probability
$$i(x, y) = \mathbb{P}(\{X = x\} \cap \{Y \leq y\}).$$
- Step 2: Construction of the conditional PMF of X, conditioned on the event $\{Y = y\}$:
$$i_y(x) = \frac{1}{f_Y(y)} \cdot \frac{d}{dy} i(x, y), \qquad x \in \mathbb{R}.$$
- Step 3: The quantity $\mathbb{E}[X|\{Y = y\}]$ is defined as the expectation with respect to the conditional PMF $i_y(x)$, i.e.,
$$\mathbb{E}[X|\{Y = y\}] := \sum_{x \in \mathbb{R}} x \cdot i_y(x).$$

Case 4: X Discrete, Y Continuous

- Step 4: Define the function $\psi_4 : \mathbb{R} \to \mathbb{R}$ as

$$\psi_4(y) \coloneqq \begin{cases} \mathbb{E}[X|\{Y = y\}], & f_Y(y) > 0, \\ 0, & f_Y(y) = 0. \end{cases}$$

- Step 5: The quantity $\mathbb{E}[X|Y]$ is simply defined as

$$\mathbb{E}[X|Y] = \psi_4(Y).$$

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Law of Iterated Expectations

Law of Iterated Expectations – 1

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X and Y be random variables w.r.t. $\mathscr{F}$.

Theorem (Law of Iterated Expectations)

Suppose that $\mathbb{E}[X]$ is well defined, i.e., not of the form $\infty - \infty$. Then,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X|Y]\big].$$

More generally, if $g : \mathbb{R} \to \mathbb{R}$ is a function such that $\mathbb{E}[g(X)]$ is well defined, then

$$\mathbb{E}[g(X)] = \mathbb{E}\big[\mathbb{E}[g(X)|Y]\big].$$

Law of Iterated Expectations – 2

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
Let X and Y be random variables w.r.t. $\mathscr{F}$.

Theorem (Law of Iterated Expectations)

If Y is a discrete random variable with PMF p_Y, then

$$\mathbb{E}[X] = \sum_{y} \mathbb{E}[X|Y=y] \cdot p_Y(y), \qquad \mathbb{E}[g(X)] = \sum_{y} \mathbb{E}[g(X)|Y=y] \cdot p_Y(y)$$

and Y is a continuous random variable with PDF f_Y, then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \mathbb{E}[X|Y=y] \cdot f_Y(y)\, dy, \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} \mathbb{E}[g(X)|Y=y] \cdot f_Y(y)\, dy$$

Miscellaneous Examples on Conditional Expectations

- Let $X \sim \text{Exponential}(1)$. Compute $\mathbb{E}[X|\{X > 1\}]$.

Miscellaneous Examples on Conditional Expectations

- Let X, Y be jointly distributed uniformly over the triangle with vertices at $(0, 0)$, $(1, 0)$, and $(0, 2)$. Compute $\mathbb{E}[X|\{Y > 1\}]$.